XMBA 201A – Managerial Economics

Office Hours Practice Problems: Block 1+2 Review

GSI: Simoni Jain | Date: Oct 14, 2026

Question 1: Decisions under uncertainty

Lectures 1 and 2

Fall 2018 Midterm

The Oregon Shakespeare Festival in Ashland has been heavily affected by smoke from the recent wildfires in Oregon and California.[1] Bill Rauch, the festival's artistic director, said in an interview that the financial losses are a result of over 20 performances of the three outdoor main-stage shows having been either canceled or moved to a smaller indoor theater, which has diminished ticket revenue.

A team at the festival meets each morning to assess the air quality and the forecast for the day to determine whether to cancel or move a performance indoors, Mr. Rauch said. If it seems as if the air quality will be poor, that night's outdoor performances are moved indoors. But if there is a chance of good air quality, organizers wait until the evening to decide. Shows are sometimes canceled at the last minute if air quality deteriorates during the day.

If the decision is made early in the day to move the show indoors, then revenues in the smaller venue will be $36,000. If instead the team decides to wait, then moving the show indoors will not be possible, and the show will either be canceled or not. If it is cancelled, the revenues are $0, while if it runs in the large outdoor venue then revenues are $80,000.

a. [5] Imagine a morning when the likelihood of the outdoor show being cancelled is 0.5. If Mr. Rauch wishes to maximize expected revenues, what decision should be made in the morning? What would the expected profit be? Draw a decision tree.

b. [5] On that morning Mr. Rauch has chosen to move the show indoors. What, if anything, can you conclude about Mr. Rauch's risk preferences? Explain why.

It turns out that festival goers may dislike the smaller indoor venue, depending on their tastes for indoor versus outdoor venues. The tastes of theatre goers for indoor venues are not correlated with the weather. With probability 1/3 the demand for the indoor venue will be high and revenues from moving the show indoor will be $48,000 while with probability 2/3 the demand will be low and revenues will be $30,000.

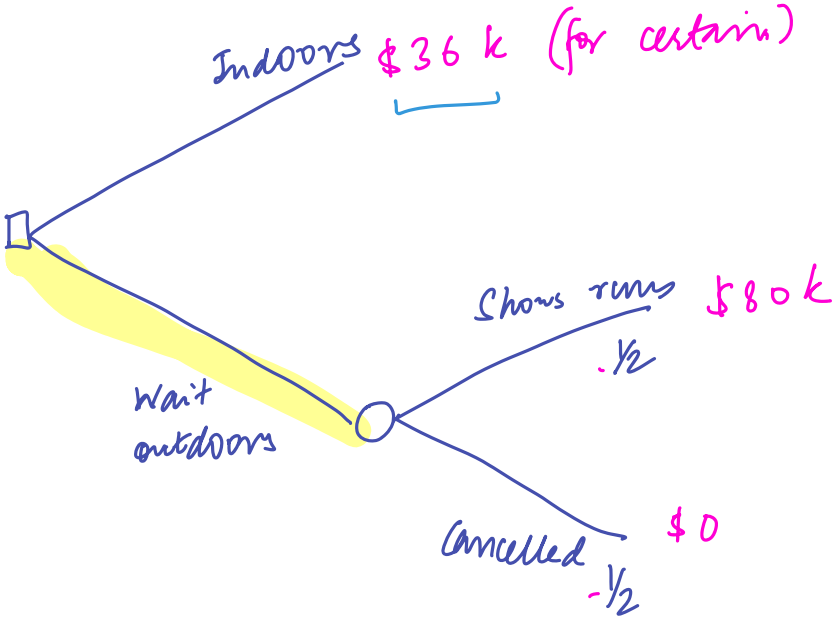
c. [10] Consider that morning when the likelihood of smoke causing a cancelation is 0.5. If Mr. Rauch is risk neutral and maximizes expected profits, which decision should he make?

d. [5] If Mr. Rauch is extremely risk averse, can you rationalize a different decision than you found in part c. above? Explain why.

e. [10] Assume Mr. Rauch is risk neutral and his assessment of the demand for indoor venues is correct (high/low with probabilities 1/3 and 2/3 respectively). Ms. Jones, a successful market researcher, knows the actual realization of the demand for indoor venues before Mr. Rauch must make his decision in the morning. How much is this information worth to Mr. Rauch? Explain.

[1] See Peter Libbey's article "Wildfire Smoke Disrupts Oregon Shakespeare Festival" New York Times, August 24, 2018.

a)



$E[R \mid \text{outdoors}] = 80 \times \frac{1}{2} = \$40k$ ✓

Swait outdoors assuming that Mrx is risk neutral!

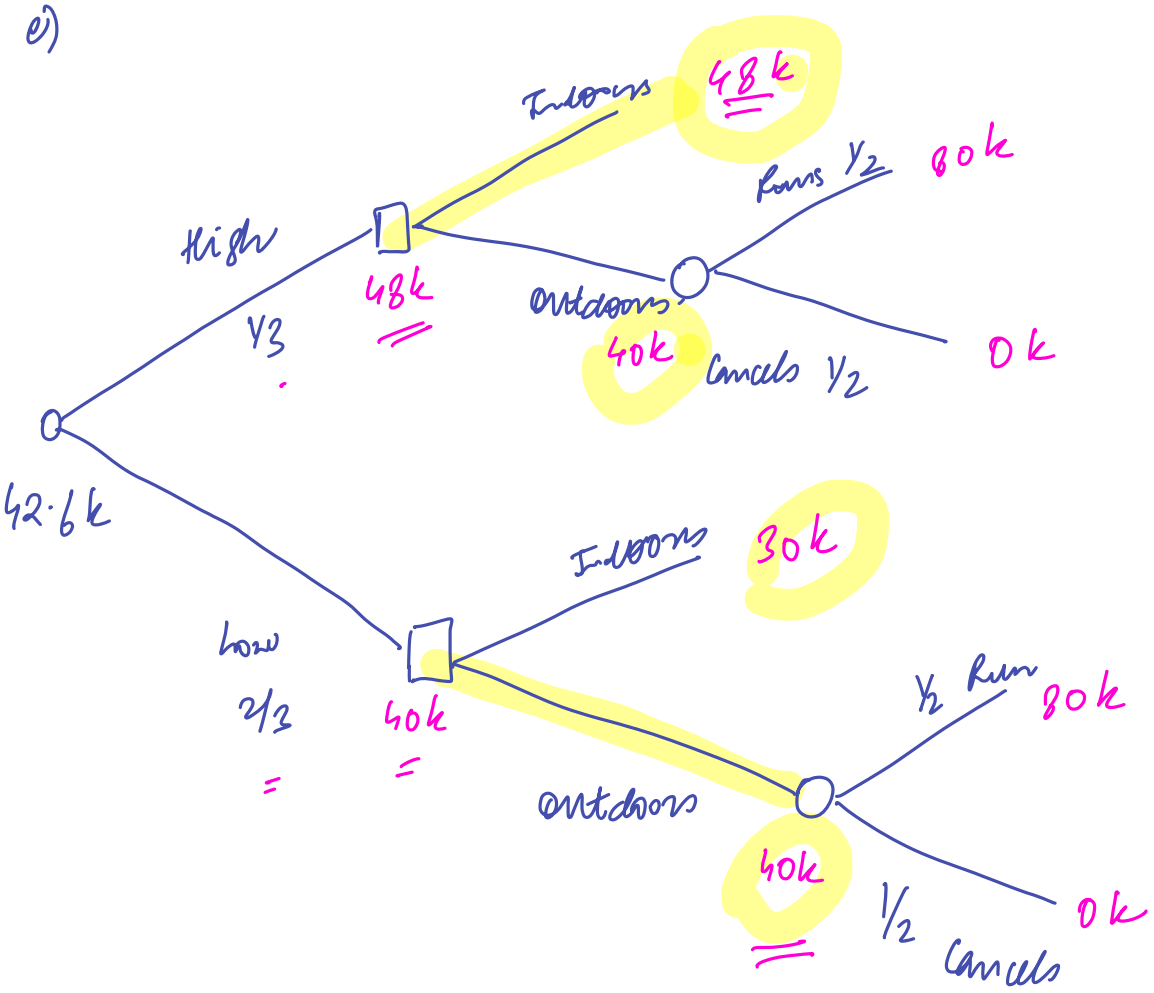
b) Must be risk-averse

c) $\frac{1}{2} \times 48k + \frac{2}{2} \times 30k = 36k$!

So $E[R \mid \text{outdoors}] > E[R \mid \text{indoors}] \Rightarrow$ stay outdoors if risk-neutral!

d) Yes, a risk-averse director could choose indoors bcoz avoiding that 50% chance of $0 payoff is worth it for him, he will be willing to give up the extra 4k that outdoors might get him, he prefers the guarantee of atleast 30k indoors.

e)



$$E[R] = 48 \times \frac{1}{3} + 40 \times \frac{2}{3}$$

$$= 16 + \frac{80}{3}$$

$$= 16 + 26.6$$

$$= 42.6\text{k}$$

Value of information = $E[R \mid \text{info}] - E[R \mid \text{without info}]$

$$= 42.6\text{k} - 40\text{k}$$

$$= \boxed{2.6\text{k}}$$

Question 2: Marginal costs pricing and capacity

Lectures 3 and 4

Fall 2022 Midterm

Haas Fine Foods is a purveyor of high-quality ingredients for local restaurants. In 2022 they specialize in foraged Morel mushrooms and authentic gelato.

On the mushroom side of the business, Robin is the only forager skilled enough to spot them. Robin is willing to work up to 40 hours a week foraging mushrooms, but not more. Robin's wage is $30/hour, and each container of mushrooms Robin harvests can be sold for $70. However, the more Robin forages, the more Robin has to search for the next mushroom.

The first ten containers Robin produces take an hour each, after which Robin gets tired, and needs 2 hours per container for the next five containers. After that Robin needs 4 hours per container so they can't produce more than 20 containers a week. After the weekend, Robin rejuvenates and is back to square one.

a) [10] How many containers of mushrooms should Haas Fine Foods sell per week?

On the Gelato side of the business, Haas Fine Foods pays $500 per week to lease their Gelato equipment. Their labor and material costs come to $4 per gallon of gelato. Demand for their Gelato is given by

$$P(Q) = 100 - 2*Q$$

b) [10] How many gallons of Gelato should Haas fine foods produce per week? At what price should they sell it to maximize profits?

c) [5] What is the elasticity of demand for Gelato at this price?

There is a fire at the HFF facility, and HFF is forced to operate with a single walk-in refrigerator for the duration of the repairs. This means they can only move a total of 20 units per week, where a "unit" is a container of mushrooms or a gallon of gelato. For example, they would be at capacity selling 11 containers of mushrooms and 9 gallons of gelato.

d) [20] (harder) How will this affect their optimal price for gelato, and how much will they sell?

a)

Containers	Hrs. per container	MC = wage × hrs.	(P − MC) = 70 − MC
1–10	1	$30	$40
11–15	2	$60	$10
16–20	4	$120	$ −50

Produce 15 containers, 16th one gets you −ve profits!

MR = MC!

b) $P(Q) = 100 - 2Q$

$MR(Q) = 100 - 4Q$ ✓

$MC = \$4$.

$MR = MC \Rightarrow 4 = 100 - 4Q$

$\Rightarrow Q^* = 24$

$P^* = 100 - 2.24$

$= 100 - 48$

$= \$52$.

$MR = MC$

$P = a - bQ$

$MR = a - 2bQ$

$TR = P.Q$

$= (a - bQ)Q$

$= aQ - bQ^2$

$MR = \frac{\partial TR}{\partial Q} = \frac{\partial}{\partial Q}(aQ - bQ^2)$

$= a - 2bQ$

d) 15 mushroom + 24 gelato > 20 units

In 1st mushroom bucket: MC for gelato: $4 + $40

$100 - 4Q = 4$

MR MC

$Q = \frac{100 - 4}{4} = 15$

14 gelato + 6 mushroom = 20 units. ✓

2nd bucket of mushrooms: 11 - 15 mushroom, 5 - 9 gelato

Gelato: MR - MC = (100 - 4Q) - 4 = 96 - 4Q

96 - 4 × 5 = 96 - 20 = 76

96 - 4 × (9) = 96 - 36 = 60 //

So if I sell 9 gelatos, I'll make $60.

With 9 gelatos, I can have 11 mushrooms (9+11=20)

But on that 11th mushroom I make $10

5th - $76 (5-9)

6th :

7th :

9th : $60